PROJECT JARVIS: PRIVATE AUTONOMOUS AI SYSTEM

Architect: Ganesh Nair | **Classification:** Personal Single-User Autonomous OS | **Edition:** Master Architecture

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1. Executive Vision: The Private 'Iron Man' Operating System

JARVIS is architected as an autonomous personal operating system dedicated exclusively to its principal, Ganesh Nair. Unlike generic passive conversational chatbots, JARVIS is a proactive, multi-agent cognitive system that initiates actions, monitors opportunities (internships, hackathons, job matches), manages daily academic assignments, coordinates across multiple AI tools, and operates desktop and browser applications under strict, zero-data-leak cryptographic boundaries.

2. Operational Core: Putting Cloned Powers into Live Action

Cloned repositories and functional libraries in _upstream/ and jarvis/powers/ are not ornamental shelfware; they are dynamically bound into the central LangGraph cognitive loop via typed schemas and executable tools:

Domain Subsystem	Active Upstream Engine	Operational Power in JARVIS
Browser & Form Filling	Playwright + DOM Extractor	Autonomous multi-step job application, form entry, and screenshot audit.
Hybrid Search	llm-search + FTS5 BM25 + RRF	Reciprocal Rank Fusion (k=60) searching local notes, code, and documents in <5ms.
Dynamic Tool Binder	ActionWeaver Type Inference	Auto-generates JSON tool schemas from Python signatures with a 12-tool VRAM budget.
Computer Control	Desktop Worker + OS Automation	Direct window focus, keystrokes, and file staging across Windows apps.
Proactive Radar	Background Cron Scheduler	Daily morning opportunity scan (jobs, deadlines, hackathons) without prompting.

3. The 8-Node Cognitive Architecture

JARVIS implements the continuous, idempotent 8-stage cognitive state machine:

- **1. Perceive:** Ingests user voice, text, background cron triggers, or desktop events.
- **2. Recall:** Hybrid SQLite FTS5 BM25 + vector search with strict tenant isolation and zero hallucinations.
- **3. Plan:** Decomposes multi-step goals into executable action steps within a 10-step ceiling.
- **4. Policy Gate:** Evaluates privilege levels (Tiers 0-3). Denies destructive actions unless signed with HMAC-SHA256 tokens.
- **5. Act:** Executes concrete powers (Playwright, desktop automation, LLM mediator).
- **6. Verify:** Validates action outputs against evidence before accepting results.
- **7. Reflect:** Records successes, isolates errors, and adjusts strategy on failure.

- **8. Remember:** Commits canonical episodic facts to encrypted SQLite WAL storage.

4. Private Multi-LLM Mediation ('Wide-Coder' Engine)

JARVIS serves as an autonomous mediator across multiple AI tools and local models. When a complex coding task or project is requested, JARVIS formulates the prompt, queries research models, pipes outputs into Antigravity or local code synthesizers, executes and debugs tests, and iterates until the solution is proven.

5. Hardware Budget & Zero-Data-Leak Privacy

- **Target Workstation:** NVIDIA GeForce RTX 4050 Laptop GPU (6GB VRAM), 24GB RAM, Windows 11.
- **Drive Protection:** C: drive restricted due to 2.56GB free space; all local model weights routed to D:\ollama\models (115GB free).
- **Zero Cloud Leaks:** 100% private single-user plan. No credentials, tokens, or personal facts leave the machine.
- **Hardware Emergency Stop:** Global 50ms kill switch shortcut (Ctrl+Alt+Shift+K) halts all running workers.

Authored for Ganesh Nair // Project JARVIS Autonomous OS Master Architecture